

MATH3026/4022 Lecture 14

Time Series using the Frequency Domain II

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- ▶ Recap of the spectral density function – definition and features.
- ▶ Writing the spectral density function in terms of the autocovariance function.
- ▶ Sample estimation of the spectral density function (the periodogram).
- ▶ Link between the autocovariance generating function and the spectral density function.
- ▶ Examples:
 - ▶ White noise process;
 - ▶ $AR(1)$ and $AR(2)$ processes;
 - ▶ $MA(1)$ process.

During Lecture 13, we introduced the concept of expressing a time series and its behaviour using the **frequency domain**.

We derived and introduced the **spectral density function** (or spectral density) for a (weakly) stationary process $\{X_t\}$, defined $g(w)$.

The spectral density function is a function of frequencies w ($w \in (-\pi, \pi]$) that determines the contribution of these frequencies to the overall variability in the process $\{X_t\}$.

Lecture 13 - Recap

We saw that the relationship between the spectral density function, $g(w)$, and the autocovariance function, $\gamma(h)$, (for a (weakly) stationary process $\{X_t\}$) is such that

$$\gamma(h) = \int_{-\pi}^{\pi} g(w) e^{-iwh} dw.$$

It follows that the variance of the process $\{X_t\}$, denoted σ_X^2 , may be evaluated as

$$\sigma_X^2 = \gamma(0) = \int_{-\pi}^{\pi} g(w) dw.$$

As noted during Lecture 13, the spectral density function and autocovariance function contain equivalent information about the process $\{X_t\}$.

Spectral Density Function

In the previous definitions and working, we have the autocovariance function, $\gamma(h)$, expressed as a function of the spectral density, $g(w)$.

Suppose that, instead, we would like to determine the spectral density function for a process, given the autocovariance function $\gamma(h)$.

In fact, if $\sum_{h=0}^{\infty} |\gamma(h)| < \infty$ then $g(w)$ always exists and is given by

$$g(w) = \frac{1}{2\pi} \sum_{h=-\infty}^{\infty} \gamma(h) e^{iwh}.$$

We shall see how to derive this result using a Fourier expansion.

Fourier Expansion of $g(w)$

First, we note that spectral density representation of a time series shares obvious similarities with the Fourier expansion of a function.

Since $g(w)$ is a function on $(-\pi, \pi]$, the Fourier series representation of $g(w)$ is

$$g(w) = \sum_{n=-\infty}^{\infty} c_n e^{inw}$$

$$\text{where } c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} g(w) e^{-inw} dw.$$

Fourier Expansion of $g(w)$

We demonstrate this result and derive the form of $g(w)$

Fourier Expansion of $g(w)$

Functional form of $g(w)$

We see that

$$g(w) = \frac{1}{2\pi} \sum_{h=-\infty}^{\infty} \gamma(h) e^{iwh}$$

Also, because $\gamma(h) = \gamma(-h)$ it follows that

$$\gamma(h) = \int_{-\pi}^{\pi} g(w) e^{iwh} dw$$

$$\text{and } g(w) = \frac{1}{2\pi} \sum_{h=-\infty}^{\infty} \gamma(h) e^{-iwh}.$$

Functional form of $g(w)$

Furthermore we can deduce that

$$g(w) = \frac{\gamma(0)}{2\pi} + \frac{1}{\pi} \sum_{h=1}^{\infty} \gamma(h) \cos(wh).$$

Note: Functional form of $g(w)$

We have seen that

$$g(w) = \frac{1}{2\pi} \sum_{h=-\infty}^{\infty} \gamma(h) e^{iwh}.$$

We note that some books/authors may put the $\frac{1}{2\pi}$ constant in different places, for example, by writing $g(w)$ as

$$\gamma(h) = \int_{-\frac{1}{2}}^{\frac{1}{2}} g(w) e^{-i2\pi wh} dw$$

where

$$g(w) = \sum_{h=-\infty}^{\infty} \gamma(h) e^{i2\pi wh}.$$

Example 14.1: White Noise Process

We know that the white noise process, $\{Z_t\}$, has the properties

$$\mathbb{E}(Z_t) = 0$$

$$\gamma(0) = \sigma_z^2 = \mathbb{E}(Z_t^2)$$

$$\gamma(h) = 0 \quad \text{for } |h| \geq 1.$$

The spectral density function for this process is

$$\begin{aligned} g(w) &= \frac{1}{2\pi} \sum_{h=-\infty}^{\infty} \gamma(h) e^{iwh} \\ &= \frac{1}{2\pi} \gamma(0) \\ &= \frac{\sigma_z^2}{2\pi}. \end{aligned}$$

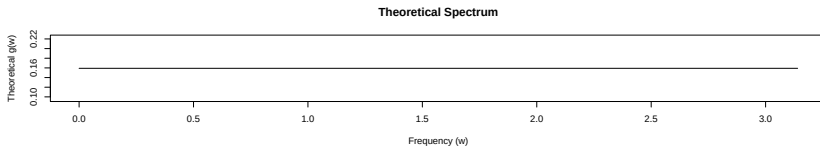
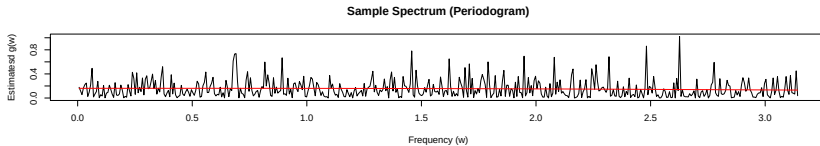
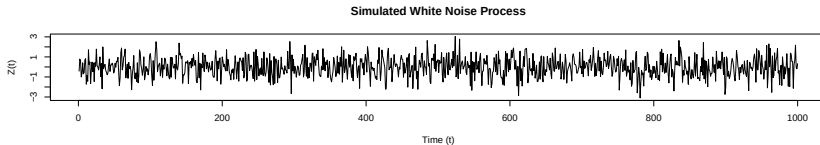
Example 14.1: White Noise Process

We see that, for the white noise process, $g(w) = \sigma_z^2/2\pi$ is constant (i.e. $g(w)$ does not depend on w).

In words, we'd interpret this as meaning that all frequencies contribute equally to the variability in the process $\{Z_t\}$.

For a white noise process, this makes sense.

Example 14.1: White Noise



The Periodogram

In Example 14.1, we saw a plot of a **sample spectral density function**.

A sample spectral density function is known as a **periodogram**.

Suppose that we have a sample of time series data $x_1, \dots, x_n$. The periodogram is a function of the frequency, w , denoted $I(w)$, where

$$\begin{aligned} I(w) &= \frac{1}{2\pi n} \left| \sum_{t=1}^n x_t e^{iwt} \right|^2 \\ &= \frac{1}{2\pi n} \left[\left\{ \sum_{t=1}^n x_t \cos(wt) \right\}^2 + \left\{ \sum_{t=1}^n x_t \sin(wt) \right\}^2 \right]. \end{aligned}$$

The Periodogram

The periodogram (sample spectral density) is related directly to the sample autocovariance function for a time series.

Recall from Lecture 6, the sample covariance between X_t and X_{t+h} (i.e. at lag h) is defined

$$c_h = \frac{1}{n} \sum_{t=1}^{n-h} (x_t - \bar{x})(x_{t+h} - \bar{x})$$

It can be shown that

$$I(w) = \frac{c_0}{2\pi} + \frac{1}{\pi} \sum_{t=1}^{\infty} c_t \cos(wt).$$

Periodogram in R

If we have a set of time series data in R, we can produce a plot of the periodogram using the command `spectrum`.

For example, if we simulate a white noise process using the command

```
x<-arima.sim(n=1000,model=list(order=c(0,0,0)),sd=1)
```

The periodogram can be obtained using the command

```
x.pgram<-spectrum(x)
```

In R the periodogram is not defined on $(-\pi, \pi]$ and should be transformed if using the notation and working from this module.

In practice, the periodogram is typically smoothed prior to plotting because $I(w)$ is not a consistent estimator of $g(w)$.

Example 14.2: $AR(1)$ Process

Recall that the (weakly) stationary $AR(1)$ process is defined

$$X_t = \phi X_{t-1} + Z_t \quad (\text{with } |\phi| < 1).$$

We know that

$$\gamma(h) = \frac{\sigma_z^2}{1 - \phi^2} \phi^{|h|} \quad \text{for } h = 0, \pm 1, \pm 2, \dots$$

We now consider the spectral density, $g(w)$, defined

$$g(w) = \frac{1}{2\pi} \sum_{h=-\infty}^{\infty} \gamma(h) e^{iwh}.$$

Example 14.2: $AR(1)$ Process

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Thus

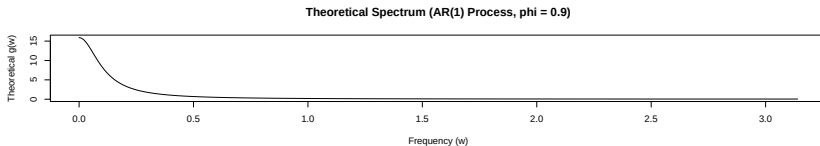
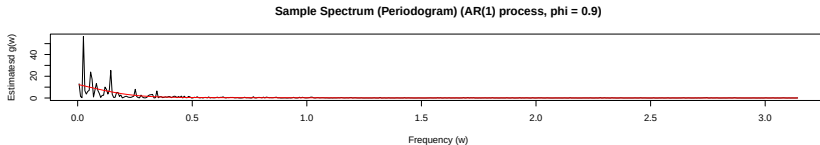
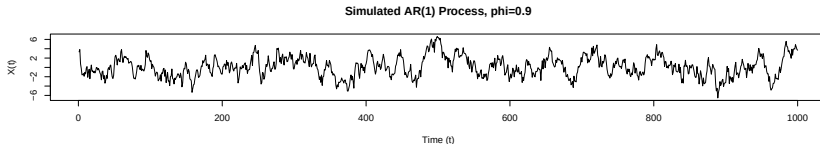
$$g(w) = \frac{\sigma_z^2}{2\pi(1 + \phi^2 - 2\phi \cos w)} \quad -\pi < w \leq \pi.$$

On the next slide, we see a time plot for a simulated $AR(1)$ time series with $\phi = 0.9$, together with a plot of the periodogram (with a smoothing spline) and the theoretical spectral density.

In this example we see that the spectral density implies that most of the contributions to the variability in the process come from low frequencies.

The concentration of the spectral density at lower frequencies implies that the process is fairly smooth and would not tend to exhibit rapid fluctuations about the mean.

Spectral Density Example: $AR(1)$ Process ($\phi = 0.9$)

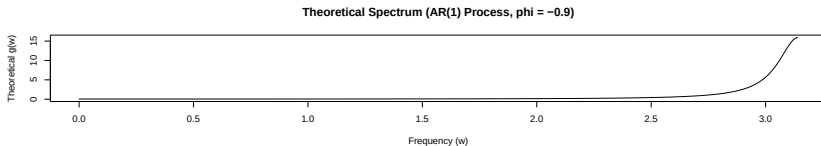
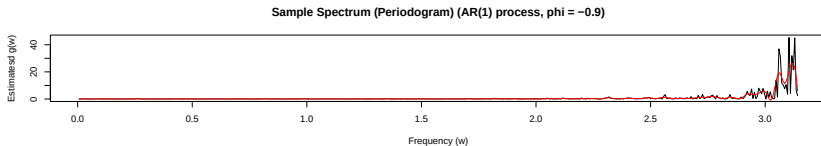
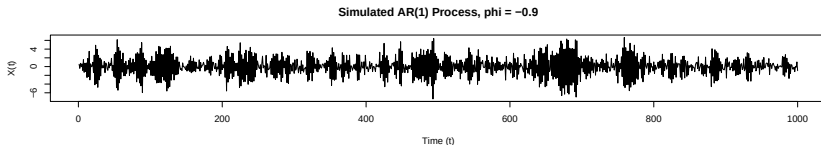


Spectral Density Example: $AR(1)$ Process ($\phi = -0.9$)

On the next slide, we see a time plot for a simulated $AR(1)$ time series with $\phi = -0.9$, together with a plot of the sample spectrum – the periodogram (with a smoothing spline) and the theoretical spectral density.

In this example we see that the spectral density implies that most of the contributions to the variability in the process come from high frequencies, thereby implying that the process is not smooth and may exhibit rapid fluctuations about the mean.

Spectral Density Example: $AR(1)$ Process ($\phi = -0.9$)



Autocovariance Generating Function

We recall the autocovariance generating function, $\Gamma(z)$

$$\begin{aligned}\Gamma(z) &= \sum_{h=-\infty}^{\infty} \gamma(h)z^h \\ &= \sigma_z^2 \psi(z)\psi(z^{-1})\end{aligned}$$

where $\psi(z)$ is derived from the $MA(\infty)$ representation of the process with

$$\begin{aligned}X_t &= \sum_{j=0}^{\infty} \psi_j Z_{t-j} \\ \psi(z) &= 1 + \psi_1 z + \psi_2 z^2 + \dots\end{aligned}$$

Autocovariance Generating Function

We know that the spectral density, $g(w)$, is written

$$\begin{aligned} g(w) &= \frac{1}{2\pi} \sum_{h=-\infty}^{\infty} \gamma(h) e^{iwh} \\ &= \frac{1}{2\pi} \Gamma(e^{iw}) \\ &= \frac{\sigma_z^2}{2\pi} \psi(e^{iw}) \psi(e^{-iw}) \end{aligned}$$

We see that we can obtain the spectral density, $g(w)$, by substituting $z = e^{iw}$ into the autocovariance generating function and dividing by 2π .

Example 14.3: $AR(2)$ process

The $AR(2)$ process is written

$$\begin{aligned}\phi(B)X_t &= Z_t \text{ where } \phi(B) = 1 - \phi_1 B - \phi_2 B^2. \\ \implies X_t &= \phi(B)^{-1} Z_t \text{ (in } MA(\infty) \text{ representation)} \\ \implies \psi(B) &= \phi(B)^{-1}\end{aligned}$$

Then

$$\begin{aligned}g(w) &= \frac{\sigma_z^2}{2\pi} \psi(e^{iw}) \psi(e^{-iw}) \\ &= \frac{\sigma_z^2}{2\pi} \phi(e^{iw})^{-1} \phi(e^{-iw})^{-1} \\ &= \frac{\sigma_z^2}{2\pi} \left[\frac{1}{(1 - \phi_1 e^{iw} - \phi_2 e^{2iw})(1 - \phi_1 e^{-iw} - \phi_2 e^{-2iw})} \right] \\ &= \frac{\sigma_z^2}{2\pi} \left[\frac{1}{1 + \phi_1^2 + \phi_2^2 - 2\phi_1(1 - \phi_2) \cos w - 2\phi_2 \cos 2w} \right].\end{aligned}$$

Example 14.3: $AR(2)$ Process

Recall that the $AR(2)$ process is written

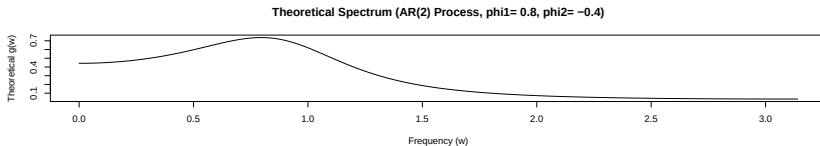
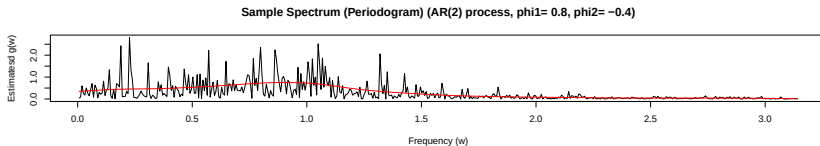
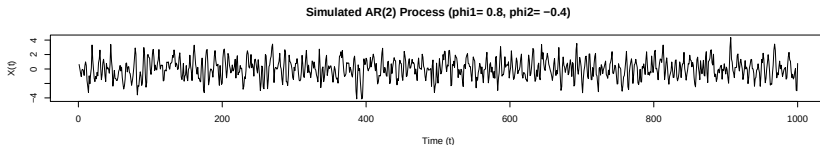
$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + Z_t.$$

If $\phi_1^2 + 4\phi_2 < 0$, we have complex roots, and the theoretical spectral density for the $AR(2)$ process is given by

$$g(w) = \frac{\sigma_z^2}{2\pi} [1 + \phi_1^2 + \phi_2^2 - 2\phi_1(1 - \phi_2)\cos w - 2\phi_2\cos 2w]^{-1}.$$

On the next slide, we see a time plot for a simulated $AR(2)$ time series with $\phi_1 = 0.8$, $\phi_2 = -0.4$, together with a plot of the sample spectrum (with a smoothing spline) and the theoretical spectral density.

Example 14.3: $AR(2)$ Process ($\phi_1 = 0.8, \phi_2 = -0.4$)



Spectral Density Function: $ARMA(p, q)$ Process

Recall that the $ARMA(p, q)$ process is defined

$$X_t = \phi_1 X_{t-1} + \dots + \phi_p X_{t-p} + Z_t + \theta_1 Z_{t-1} + \dots + \theta_q Z_{t-q}$$

which can be re-written

$$\phi(B)X_t = \theta(B)Z_t$$

where

$$\phi(B) = 1 - \phi_1 B - \dots - \phi_p B^p;$$

$$\theta(B) = 1 + \theta_1 B + \dots + \theta_q B^q.$$

Spectral Density Function: $ARMA(p, q)$ Process

It follows that

$$X_t = \frac{\theta(B)}{\phi(B)} Z_t.$$

It can be shown that the spectral density function for the $ARMA(p, q)$ process is given by

$$\begin{aligned} g(w) &= \frac{\sigma_z^2}{2\pi} \frac{\theta(e^{iw})\theta(e^{-iw})}{\phi(e^{iw})\phi(e^{-iw})} \\ &= \frac{\sigma_z^2}{2\pi} \frac{|\theta(e^{iw})|^2}{|\phi(e^{iw})|^2}. \end{aligned}$$

Example 14.4: $MA(1)$ process

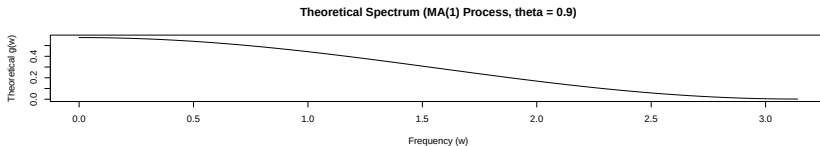
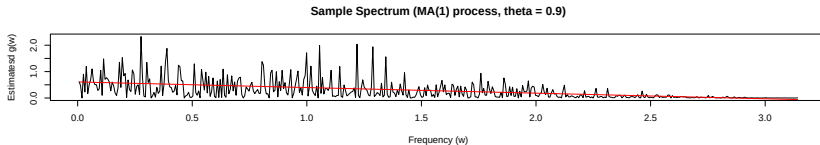
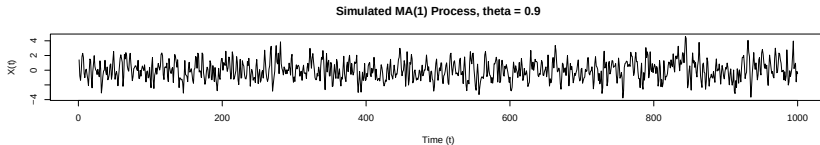
Example 14.4: $MA(1)$ process

We see that

$$g(w) = \frac{\sigma_z^2}{2\pi} [1 + \theta^2 + 2\theta \cos w] .$$

On the next slide, we see a time plot for a simulated $MA(1)$ time series with $\theta = 0.9$, together with a plot of the periodogram (with a smoothing spline) and the theoretical spectral density.

Example 14.4: $MA(1)$ process



Example 14.4: $MA(1)$ process

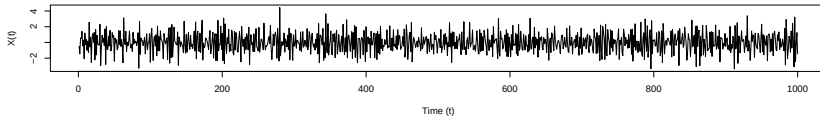
To interpret the spectral density function, we note that $g(w)$ is a decreasing function of w , thereby implying that lower frequencies have most influence on the behaviour of the process.

The process will not tend to oscillate rapidly and there may be gradual, long-range, fluctuations in the process.

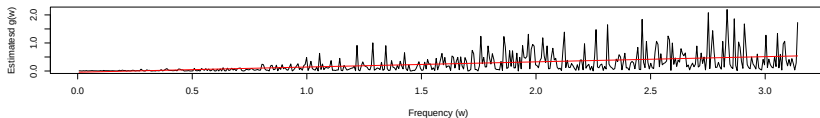
Now we consider similar plots, but for an $MA(1)$ process where $\theta = -0.8$.

Example 14.4: $MA(1)$ process

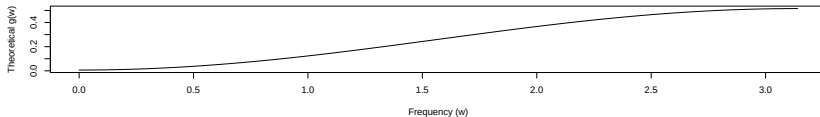
Simulated $MA(1)$ Process, $\theta = -0.8$



Sample Spectrum ($MA(1)$ process, $\theta = -0.8$)



Theoretical Spectrum ($MA(1)$ Process, $\theta = -0.8$)



Example 14.4: $MA(1)$ process

We note that $g(w)$ is an increasing function of w , thereby implying that higher frequencies have most influence on the behaviour of the process.

The process will tend to oscillate rapidly and exhibit fewer long-range fluctuations.

This makes sense when we consider the ACF of the $MA(1)$ process.

Example 14.4: $MA(1)$ process

From Lecture 3, recall that the autocorrelation function for the $MA(1)$ process is

$$\rho(h) = \begin{cases} 1 & h = 0; \\ \frac{\theta}{1 + \theta^2} & h \in \{-1, +1\}; \\ 0 & \text{otherwise.} \end{cases}$$

When θ is negative, we expect negative linear correlation between X_t and X_{t+1} .

Negative correlation implies that a *large* value of X_t is likely to be followed by a *small* value of X_{t+1} (and vice versa).

This suggests that the process is likely to oscillate in the short term (i.e. high frequencies contribute most to the spectral density function).

Example 14.4: $MA(1)$ process

Conversely, when θ is positive, we expect positive linear correlation between X_t and X_{t+1} .

Positive correlation implies that a *large* value of X_t is likely to be followed by another *large* value of X_{t+1} (likewise for smaller values).

This suggests that the process is not likely to oscillate in the short term (i.e. high frequencies contribute least to the spectral density function).

Lecture 14: Learning Outcomes

- ▶ Be able to write the spectral density function in terms of the autocovariance function of a time series.
- ▶ Understand how the **periodogram** may be used to estimate the spectral density function.
- ▶ Understand the link between the autocovariance generating function and the spectral density function.
- ▶ Know how to calculate the spectral density function for $ARMA(p, q)$ processes, including the examples covered during this lecture.
- ▶ Be able to interpret the spectral density function in terms of how a process is likely to behave in the long and short terms.